

NT Sentinel

network-monitor - CLI network monitoring for macOS and Linux.

Project: ~/network-monitor

Config: ~/.config/new-tower/session.conf

1. Build and run

Build once, then run commands through cabal:

```
cd ~/network-monitor
cabal build
cabal run network-monitor -- menu
cabal run network-monitor -- --help
```

Optional global install:

```
cabal install network-monitor --installdir=~/.local/bin
network-monitor flow
```

2. Interactive menu

Launch the full menu with ASCII header and saved session defaults:

```
cabal run network-monitor -- menu
```

#	Feature
1	Interface Statistics
2	Live Bandwidth Monitor (Watch)
3	TCP Connections
4	Visual NetView
5	Traffic Flow (Emit Monitor)
6	Network Dashboard
7	Ping Host
8	Port Check
9	Top Remote Hosts
10	Configure Session Options
11	Help
12	Mission Control
13	App Traffic Summary
14	Traceroute
15	DNS Trace
16	Listening Ports
17	Network Intel
18	Multi-Ping Board

19	Export Session Report
0	Exit

3. Commands overview

Command	Description
interfaces (if)	Interface byte and packet counters
watch (w)	Live bandwidth monitor with spark bars
connections (conn)	Active TCP connections
dashboard (dash)	One-screen network status overview
netview (nv)	Live visual network topology map
flow (emit)	Live outbound traffic / emit monitor
trace	Traceroute to a host
dns	DNS record trace for a domain
listen	Local listening ports and owning processes
apps	Live per-app traffic summary
mission	All-in-one mission control panel
intel	Public IP, gateway, Wi-Fi, VPN snapshot
report	Export session report to ~/.config/new-tower/reports/
mping	Multi-target ping board
ping	Ping latency statistics
top	Top remote hosts by connection count
port	Check if a TCP port is open
menu	Interactive menu

4. Traffic flow (emit monitor)

Shows outbound bytes/sec per remote host, active flows, process names, service labels, alerts, DNS names, and per-app emitters.

```
cabal run network-monitor -- flow
cabal run network-monitor -- flow -I en0 -i 2
cabal run network-monitor -- flow --block bad.example --emit-alert 524288 --log
```

5. Mission control and intel

```
cabal run network-monitor -- mission
cabal run network-monitor -- intel
cabal run network-monitor -- apps
cabal run network-monitor -- report
```

6. Diagnostics

```
cabal run network-monitor -- trace -H 8.8.8.8
cabal run network-monitor -- dns -H google.com
```

```
cabal run network-monitor -- listen
cabal run network-monitor -- ping -H 1.1.1.1 --pings 4
cabal run network-monitor -- port -H google.com -p 443
cabal run network-monitor -- top
```

7. Multi-ping board

```
cabal run network-monitor -- mping -F 8.8.8.8 -F 1.1.1.1
# Favorites also load from session.conf (menu option 10, field 6)
```

8. Watch mode (continuous)

```
cabal run network-monitor -- watch -I en0
cabal run network-monitor -- watch -I en0 -i 2
cabal run network-monitor -- watch -I en0 -c 10
cabal run network-monitor -- watch
```

DOWN = incoming bytes/sec. UP = outgoing bytes/sec.

9. Interface stats

```
cabal run network-monitor -- interfaces
cabal run network-monitor -- interfaces -I en0
```

10. TCP connections

```
cabal run network-monitor -- connections
cabal run network-monitor -- connections -l 10
cabal run network-monitor -- connections -s LISTEN
```

11. Session configuration

Saved at ~/.config/new-tower/session.conf. Edit via menu option 10.

Key	Default / purpose
interface	en0 - default interface filter
interval	1.0 - refresh seconds
count	0 - auto-stop after N refreshes
state	ESTABLISHED - connection filter
limit	50 - max connection rows
favorites	8.8.8.8,1.1.1.1 - multi-ping targets
blocklist	Alert when flows match host prefix
emit_alert	524288 - high emit rate alert (B/s)
logging	true - append to sessions.log

Logs and reports:

```
~/.config/new-tower/sessions.log
~/.config/new-tower/reports/
```

12. Command-line options

Flag	Default / meaning
-i, --interval SECS	Refresh interval (default: 1.0)
-c, --count N	Refresh count; 0 = until Ctrl-C
-I, --interface IFACE	Limit to interface(s), e.g. en0
-s, --state STATE	Connection state (default: ESTABLISHED)

-l, --limit N	Max rows (default: 50)
-H, --host HOST	Target host (default: 8.8.8.8)
-p, --port PORT	Target port (default: 443)
--pings N	Ping packet count (default: 4)
-F, --favorite HOST	Favorite host for mping (repeatable)
--block HOST	Blocklisted host prefix (repeatable)
--emit-alert BPS	Emit alert threshold (default: 524288)
--log	Append session events to sessions.log
-h, --help	Show help text

13. Interface names (macOS)

```
ifconfig -l
```

Name	Typical use
en0	Wi-Fi
lo0	Loopback

14. Quick reference

Goal	Command
Interactive menu	cabal run network-monitor -- menu
Live bandwidth	cabal run network-monitor -- watch -l en0
Outbound traffic	cabal run network-monitor -- flow
Mission control	cabal run network-monitor -- mission
Network intel	cabal run network-monitor -- intel
Export report	cabal run network-monitor -- report
Help	cabal run network-monitor -- --help